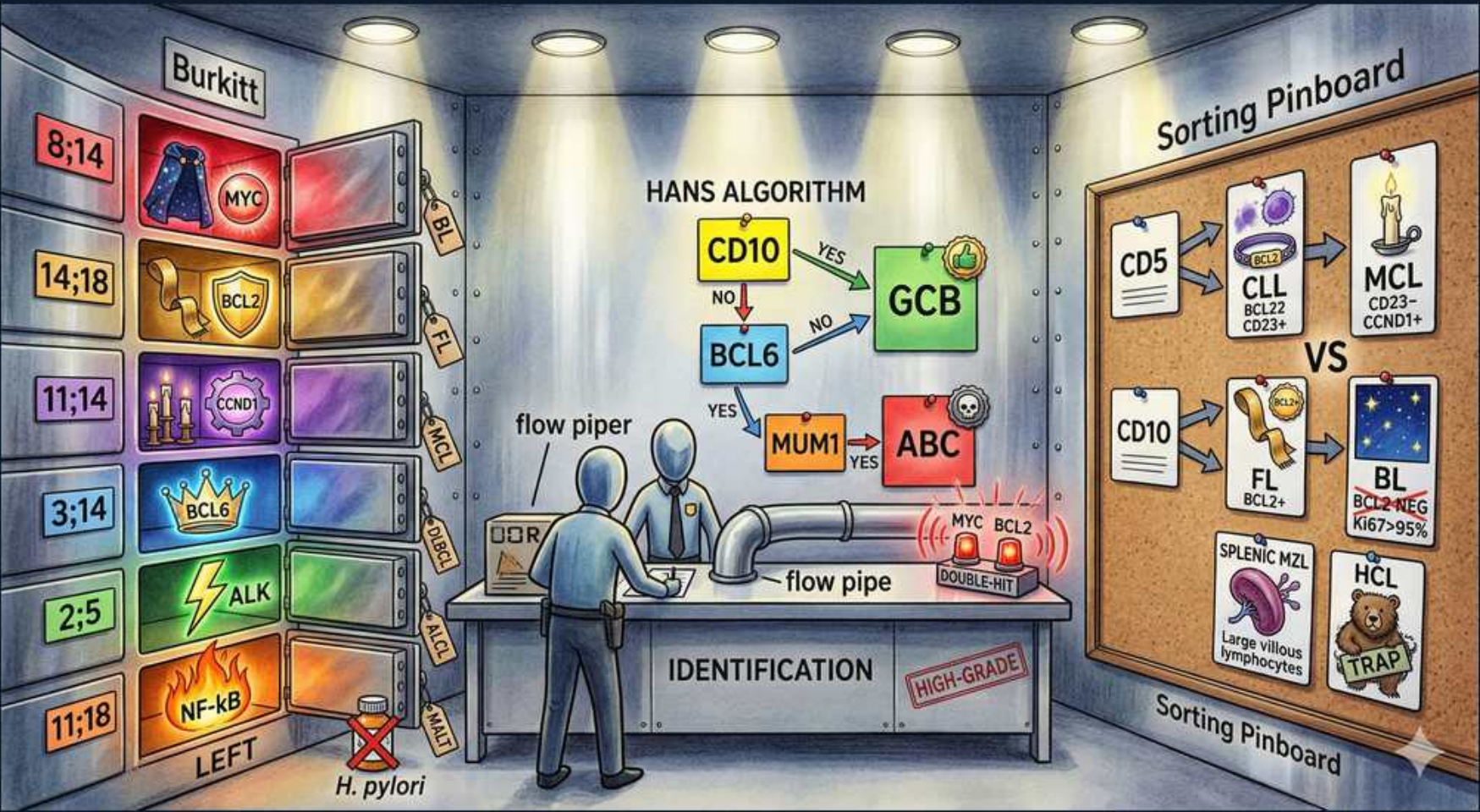


NHL — Translocation Vault (bank vault)



1. t(8;14) MYC — Burkitt

WHAT YOU SEE

'8;14' vault box with 'MYC' bat, labeled Burkitt

WHAT IT ENCODES

Burkitt lymphoma is defined by t(8;14)(q24;q32), which juxtaposes the c-MYC oncogene (chromosome 8) next to the immunoglobulin HEAVY-chain enhancer (chromosome 14), driving constitutive MYC transcription and the highest proliferation rate of any tumor (Ki-67 ~95-100%). Histology is the classic 'STARRY SKY' (tingible-body macrophages clearing apoptotic debris). Three forms: ENDEMIC (African, EBV-associated, jaw/facial mass), SPORADIC (abdominal/ileocecal mass), and IMMUNODEFICIENCY-associated (HIV). Its explosive growth makes TUMOR LYSIS SYNDROME the major emergency, and it requires intensive regimens (DA-EPOCH-R or hyper-CVAD) with CNS prophylaxis, not R-CHOP.

BOARD TRAP

WRONG to assign MYC to follicular lymphoma or mantle cell. The discriminator: MYC/t(8;14) = Burkitt (Ki-67 ~100%, BCL2-negative); BCL2/t(14;18) = follicular; cyclin D1/t(11;14) = mantle. Variant Burkitt translocations swap in the Ig LIGHT-chain loci: t(2;8) (kappa) and t(8;22) (lambda) — still MYC, still Burkitt.

2. t(14;18) BCL2 — Follicular

WHAT YOU SEE

'14;18' vault box with 'BCL2' shield, FL tag

WHAT IT ENCODES

Follicular lymphoma carries t(14;18)(q32;q21), placing the BCL2 anti-apoptosis gene under the Ig heavy-chain enhancer so BCL2 is overexpressed and the cells RESIST apoptosis (they accumulate rather than proliferate). It is the most common INDOLENT NHL, CD10+/BCL2+/BCL6+, with waxing-and-waning lymphadenopathy; it is generally INCURABLE but with long survival, so asymptomatic low-burden disease is often just observed ('watch and wait'). Grade 3B or transformation to DLBCL changes management to aggressive R-CHOP.

BOARD TRAP

WRONG to call BCL2 positivity by immunohistochemistry diagnostic of follicular lymphoma, or to confuse the genes. The discriminator: reactive germinal centers are BCL2-NEGATIVE while follicular lymphoma is BCL2-positive on IHC — but BCL2 PROTEIN is also expressed in many other lymphomas; the specific lesion is the t(14;18) gene rearrangement. Don't mix up BCL2 (follicular) with BCL6 (DLBCL) or BCL1/cyclin D1 (mantle).

3. t(11;14) CCND1 — Mantle

WHAT YOU SEE

'11;14' vault box with 'CCND1' crown, MCL tag

WHAT IT ENCODES

Mantle cell lymphoma is defined by t(11;14)(q13;q32) fusing CCND1 (cyclin D1, aka BCL1) to the Ig heavy-chain enhancer, causing cyclin D1 overexpression that pushes cells through the G1→S checkpoint. The cells are CD5+, CD20+, CD23-NEGATIVE, FMC7+, and nuclear cyclin D1+ (and SOX11+). It is the 'worst of both worlds' — aggressive AND incurable — and a classic cause of LYMPHOMATOUS POLYPOSIS of the GI tract. Treatment in fit patients is intensive induction (e.g., R-DHAP/hyper-CVAD) + autologous transplant; relapsed disease responds to BTK inhibitors (ibrutinib, acalabrutinib) and brexucabtagene CAR-T.

BOARD TRAP

WRONG to call a CD5+ small B-cell lymphoma 'CLL' without checking cyclin D1 and CD23. The discriminator: BOTH CLL and mantle cell are CD5+, but CLL is CD23-POSITIVE/cyclin D1-negative, whereas mantle cell is CD23-NEGATIVE/cyclin D1-positive with t(11;14) — getting this wrong sends an aggressive, transplant-eligible MCL down a 'watch and wait' CLL pathway.

4. t(3;14) BCL6 — DLBCL

WHAT YOU SEE

'3;14' vault box with 'BCL6'

WHAT IT ENCODES

BCL6 (chromosome 3q27) rearrangements — t(3;14) and variants — are the most common structural lesion in diffuse large B-cell lymphoma; BCL6 is a germinal-center master transcriptional repressor. DLBCL is the most common NHL overall and the most common aggressive B-cell lymphoma, curable in ~60% with R-CHOP. Critically, when MYC is rearranged TOGETHER with BCL2 (and/or BCL6), the tumor becomes a 'DOUBLE-HIT' / high-grade B-cell lymphoma with a much worse prognosis.

BOARD TRAP

WRONG to equate a high MYC-protein or BCL2-protein level (by IHC) with a true 'double-hit' lymphoma. The discriminator: 'double-hit'/'double-expressor' are different — double-HIT requires concurrent MYC AND BCL2/BCL6 chromosomal REARRANGEMENTS proven by FISH (a genetic event, high-grade WHO category), whereas 'double-EXPRESSOR' is merely protein overexpression by IHC. Only true double-hit mandates intensified therapy like DA-EPOCH-R.

5. t(2;5) ALK — ALCL

WHAT YOU SEE

'2;5' vault box with 'ALK' bolt

WHAT IT ENCODES

Anaplastic large cell lymphoma (the one T-cell tumor in this vault) carries t(2;5)(p23;q35), generating the NPM1-ALK fusion that constitutively activates the ALK tyrosine kinase. The cells are uniformly CD30-positive with 'hallmark' horseshoe-nuclei cells. ALK-POSITIVE ALCL occurs in children/young adults and has a GOOD prognosis (~70-80% 5-year survival); ALK-negative ALCL is older and intermediate. Frontline is now brentuximab vedotin (anti-CD30) + CHP per ECHELON-2.

BOARD TRAP

WRONG to treat ALK positivity as an adverse marker the way it is in lung cancer. The discriminator: in ALCL, ALK-POSITIVE = FAVORABLE (curable with chemo); ALK-NEGATIVE = worse prognosis — the opposite intuition from solid tumors. Also don't confuse this CD30+ T-cell lymphoma with Hodgkin lymphoma, which is also CD30+ but is CD15+/CD20-variable with Reed-Sternberg cells.

6. t(11;18) NF-kB — MALT

WHAT YOU SEE

'11;18' vault box with 'NF-kB' flame

WHAT IT ENCODES

Extranodal marginal-zone (MALT) lymphoma can carry t(11;18)(q21;q21), creating the API2-MALT1 fusion that constitutively activates NF-κB signaling, freeing the tumor from its antigen-driven (H. pylori) growth dependence. MALT lymphomas arise at sites of chronic inflammation: stomach (H. pylori), salivary gland (Sjögren), thyroid (Hashimoto), lung, ocular adnexa (Chlamydia psittaci), and small bowel (Campylobacter, 'IPSID'). t(11;18) is the single most important predictor that antibiotic (H. pylori) eradication will FAIL.

BOARD TRAP

WRONG to keep treating a gastric MALT lymphoma with H. pylori eradication alone when it isn't regressing. The discriminator: a t(11;18)-POSITIVE gastric MALT lymphoma is antibiotic-RESISTANT (NF-κB is now autonomously driven) and needs radiotherapy or rituximab — so persistent/recurrent disease after eradication should prompt FISH for t(11;18) and a switch in strategy.

7. H. pylori hourglass

WHAT YOU SEE

Small 'H. pylori' hourglass

WHAT IT ENCODES

Gastric MALT lymphoma is the prime example of an antigen-driven, potentially CURABLE-with-antibiotics cancer: in t(11;18)-NEGATIVE, early-stage disease, H. PYLORI ERADICATION (triple therapy — a PPI + clarithromycin + amoxicillin, or bismuth quadruple therapy) induces lymphoma regression in ~60-80% of cases, often over several months. Endoscopic surveillance confirms regression; non-responders or t(11;18)+ cases get radiotherapy or rituximab. This is the classic 'treat the infection, cure the cancer' board vignette.

BOARD TRAP

WRONG to declare H. pylori eradication a failure too early or to skip confirming eradication. The discriminator: MALT regression after antibiotics can take 3-12 MONTHS, so durable endoscopic follow-up is required — and eradication success must be documented (urea breath test or stool antigen off PPI); only after confirmed eradication WITHOUT regression do you escalate to radiation/rituximab.

8. Hans algorithm CD10

WHAT YOU SEE

'CD10' yellow box, top of Hans algorithm

WHAT IT ENCODES

The Hans algorithm uses three immunohistochemistry stains — CD10, BCL6, and MUM1/IRF4 — to assign DLBCL a cell-of-origin: GCB (germinal-center B-cell) vs non-GCB (ABC). The first node is CD10: CD10-POSITIVE → GCB type, which carries a BETTER prognosis. CD10 is a germinal-center marker also expressed in follicular lymphoma and Burkitt. Cell-of-origin matters because ABC/non-GCB DLBCL is NF-κB-driven and does worse with R-CHOP.

BOARD TRAP

WRONG to think the Hans IHC algorithm is equivalent to gene-expression profiling or that it overrides genetics. The discriminator: Hans IHC is a SURROGATE for the gold-standard gene-expression cell-of-origin assignment and is imperfect; and regardless of GCB vs ABC, you still must exclude MYC/BCL2 double-HIT rearrangements by FISH, because a GCB-type DLBCL can still be a high-grade double-hit lymphoma.

9. BCL6 node

WHAT YOU SEE

'BCL6' blue box in algorithm

WHAT IT ENCODES

In the Hans algorithm, if CD10 is NEGATIVE the next branch is BCL6: a CD10-/BCL6-POSITIVE/MUM1-NEGATIVE tumor is still classified as GCB type, whereas CD10-/BCL6+/MUM1+ (or BCL6-negative) falls into the non-GCB/ABC category. BCL6 is a normal germinal-center transcriptional repressor, so its expression points toward a germinal-center origin. This branching logic separates the better-prognosis GCB from the worse-prognosis ABC subtype.

BOARD TRAP

WRONG to equate BCL6 PROTEIN expression (IHC, used in the Hans algorithm) with a BCL6 GENE rearrangement (t(3;14) translocation). The discriminator: the algorithm reads BCL6 staining as a germinal-center marker for cell-of-origin, but a BCL6 chromosomal rearrangement is a separate genetic event — and only when MYC + BCL6 (± BCL2) are co-rearranged does it become a high-grade double/triple-hit lymphoma.

10. GCB vs ABC

WHAT YOU SEE

'GCB' green and 'ABC' red endpoints

WHAT IT ENCODES

DLBCL splits by cell-of-origin into GCB (germinal-center B-cell-like) and ABC (activated B-cell-like, = non-GCB). GCB tends to carry t(14;18)/BCL2 and EZH2 mutations and has a BETTER outcome; ABC is driven by chronic active B-cell-receptor → NF-κB signaling (MYD88, CD79B mutations) and has a WORSE outcome with R-CHOP. This biology underlies trials adding NF-κB-targeting agents (ibrutinib, lenalidomide) selectively to ABC/non-GCB disease.

BOARD TRAP

WRONG to assume cell-of-origin alone dictates frontline therapy outside a trial. The discriminator: although ABC/non-GCB has a worse prognosis, R-CHOP remains the standard frontline for both subtypes (cell-of-origin-tailored therapy has not beaten it in practice) — the actionable molecular flag that DOES change frontline therapy is MYC/BCL2 double-HIT, which moves you to DA-EPOCH-R.

11. Double-hit MYC+BCL2

WHAT YOU SEE

'MYC BCL2 DOUBLE-HIT' alarm

WHAT IT ENCODES

'Double-hit' (and 'triple-hit') lymphoma = high-grade B-cell lymphoma with concurrent MYC rearrangement PLUS BCL2 and/or BCL6 rearrangement (proven by FISH), now its own WHO category ('HGBL with MYC and BCL2 rearrangements'). It is very aggressive with frequent CNS involvement and a poor outcome with standard R-CHOP, so it is treated with intensified regimens such as DA-EPOCH-R plus CNS prophylaxis. Suspect it in any high-grade-appearing GCB DLBCL with very high Ki-67.

BOARD TRAP

WRONG to give standard R-CHOP to a double-hit lymphoma, or to call any MYC+/BCL2+ tumor by IHC 'double-hit.' The discriminator: double-HIT is defined by genetic REARRANGEMENTS on FISH (not protein on IHC, which is 'double-expressor'), and it specifically requires intensified therapy (DA-EPOCH-R) with CNS-directed prophylaxis — under-staging it as ordinary DLBCL undertreats a curable-but-aggressive disease.

12. CD5 → CLL vs MCL

WHAT YOU SEE

'CD5' branch to CLL (CD23+) and MCL (CD23-/CCND1+)

WHAT IT ENCODES

CD5 is aberrantly expressed on two mature B-cell neoplasms: CLL/SLL and MANTLE CELL lymphoma. The tie-breaker is CD23 and cyclin D1: CLL is CD5+/CD23-POSITIVE/cyclin D1-negative (and dim CD20, dim surface Ig, LEF1+), whereas mantle cell is CD5+/CD23-NEGATIVE/cyclin D1-POSITIVE with t(11;14). CLL also shows characteristic 'smudge cells' on smear; mantle cell can present with GI lymphomatous polyposis. Getting the split right is high-yield because management and prognosis diverge sharply.

BOARD TRAP

WRONG to stop at 'CD5+ small B-cell lymphoma = CLL.' The discriminator: a CD5+ lymphoma that is CD23-NEGATIVE and cyclin D1-POSITIVE with t(11;14) is mantle cell — aggressive and transplant-eligible — NOT indolent CLL; missing this distinction (CD23 and cyclin D1 are the decisive stains) leads to dangerously inadequate management.

13. CD10 → FL vs Burkitt

WHAT YOU SEE

'CD10' branch to FL (BCL2+) and BL (BCL2- Ki67 ~95-100%)

WHAT IT ENCODES

CD10 marks germinal-center-derived B-cell tumors; among CD10-positive lymphomas, FOLLICULAR (BCL2-POSITIVE, t(14;18), LOW Ki-67, indolent) is separated from BURKITT (BCL2-NEGATIVE, t(8;14)/MYC, Ki-67 ~95-100%, 'starry sky,' explosively aggressive). The proliferation index is the fastest discriminator: a near-100% Ki-67 essentially excludes follicular and points to Burkitt or high-grade transformation. Both are CD10+/BCL6+, so BCL2 status and Ki-67 do the work.

BOARD TRAP

WRONG to read 'CD10-positive germinal-center tumor' as automatically follicular/indolent. The discriminator: a CD10+ tumor that is BCL2-NEGATIVE with Ki-67 ~100% and a starry-sky pattern is BURKITT — an oncologic EMERGENCY needing intensive therapy and tumor-lysis prophylaxis — whereas follicular is BCL2-positive with low Ki-67 and is indolent; confusing the two delays life-saving treatment.

14. Splenic MZL / HCL

WHAT YOU SEE

'Splenic MZL — large villous lymphocytes' and 'HCL TRAP'

WHAT IT ENCODES

Two splenomegaly-predominant small B-cell neoplasms to contrast: SPLENIC MARGINAL-ZONE lymphoma shows circulating 'villous lymphocytes' (polar cytoplasmic projections), is often associated with HEPATITIS C, and may respond to antiviral therapy. HAIRY CELL LEUKEMIA shows cells with circumferential 'hairy' projections, is TRAP-positive (and CD11c/CD25/CD103/CD123+, annexin A1+), carries the BRAF V600E mutation, causes a 'dry tap' marrow with reticulin fibrosis, MONOCYTOPENIA, and massive splenomegaly, and is exquisitely sensitive to purine analogs (cladribine/pentostatin); refractory disease responds to BRAF inhibitors (vemurafenib).

BOARD TRAP

WRONG to lump 'splenomegaly + cytopenias + atypical lymphocytes' into one diagnosis. The discriminator: hairy cell leukemia is TRAP+/BRAF V600E+ with MONOCYTOPENIA and a dry-tap fibrotic marrow (treat with cladribine), whereas splenic marginal-zone lymphoma has villous lymphocytes and a hepatitis-C link (treat the HCV) — the BRAF mutation and monocytopenia are the clinching features of hairy cell.